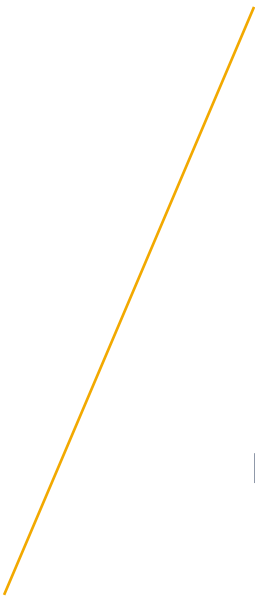


PANDROL

MANUFACTURING PRODUCT SPECIFICATION

Pandrol Anchor Components



LW-L

06 June 2023

Partners in excellence



Revision History

Version	Date	Authorised by	Comments
1	11 March 2019	Head of Engineering and Support	New issue. Mod 3354
2	29 March 2021	Head of Engineering and Support	Square headed bolts and screws added and correction to standard and Table 4 for Bolts with Trapezoidal threads ECR15386
3	06 June 2023	Head of Engineering and Support	Annex A added to Cover Proof-Load Testing Of Nut; Steel #45 added to Table 28. ECR15695

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General note

The manufacturer shall check that specification reference and issue code on this document corresponds exactly to the code on purchase order.

The use of equivalent International Standards to those specified below is subject to written agreement from Pandrol Rail Fastening Systems, Product Line.

The equipment used during these tests shall be included in a recognised calibration system and shall be traceable to International Standards.

Manufacturers shall inform the owner of the tooling of any requests made by other purchasing companies to use the tooling.

Any sub-contracted production or service shall be disclosed to the purchasing company. Written approval is required.

For packaging, labelling and shipping requirements refer to the purchase order.

1. Scope

To specify the manufacturing and inspection requirements for Pandrol brand anchor components.

This document does not include requirements for threaded inserts / dowels.

2. Raw materials, processes and coatings

Raw materials, manufacturing processes and corrosion preventive coatings (where specified) shall be in accordance with the relevant product requirements tables in Section 7 to Section 17 of this document.

3. Pre-production (initial sample) approval

3.1. Applicability

This applies:

- to all new suppliers
- to all new or modified tooling facilities
- to all new or modified production facilities
- when required

Where the purchasing company has already approved production, further pre-production samples may not be required.

Manufacturers shall inform the owner of the tooling of any requests made by other purchasing companies to use the tooling.

3.2. Requirements

Manufacturers shall ensure their products meet all relevant requirements of this specification by testing three samples from each tool in accordance with the requirements stated in the relevant component specific requirements in Section 7 to Section 17.

If all requirements are met satisfactorily, samples and test results shall be submitted to Pandrol for their approval.

If the tests on the samples are approved by Pandrol, bulk production can begin.

4. Bulk production

4.1. Applicability

This section applies only to anchor component manufacturers who have received written pre-production (initial sample) approval from Pandrol.

4.2. Requirement

4.2.1. New tooling

For all new tooling pre-production samples shall be tested in accordance with in the relevant component specific requirements in Section 7 to Section 17.

4.2.2. Applicable changes

Any changes that may affect the anchor components and previous approvals shall be notified in writing to Pandrol who may require an inspection report, production or pre-production samples.

4.2.3. Inspection

The specific test requirements are presented individually by component type in Section 7 to Section 17 and shall be met routinely during bulk manufacture.

The tests detailed are the responsibility of the anchor component manufacturer and shall be carried out by the anchor component manufacturer or a mutually acceptable independent testing authority.

Pandrol may undertake or repeat all tests detailed.

All test results shall be treated as though they were independently obtained for each property.

4.3. Acceptance

Each batch shall be accepted if the samples meet all the requirements of this specification. Anchor components shall not be dispatched until the purchasing company has given written approval.

4.4. Failure

In the event of a failure the test which failed shall be repeated by taking further samples from the rejected production batch at the same inspection frequency.

If the repeat test fails, the entire batch shall be rejected unless 100% inspection can be undertaken and the unacceptable items in the batch removed.

5. Records

All quality records shall be retained by the anchor component manufacturer for a minimum of seven years.

6. Certification according to drawing and specification

The anchor component manufacturer shall submit with each batch of anchor components (or order if agreed) a certificate of conformity and associated test results.

7. Hexagon Headed Bolts and Screws

7.1. Requirements

Pandrol hexagon headed bolts and screws conform to the requirements of ISO 4014 and Table 1.

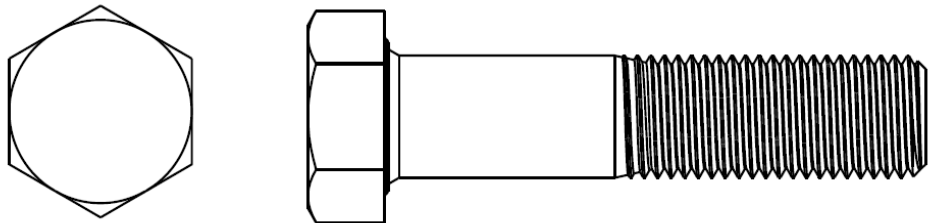


Figure 1 – Hexagon Headed Bolt

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6g		
	International Standard	ISO 724, ISO 965-1		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 1 – Product requirements for Hexagon Headed Bolts and Screws

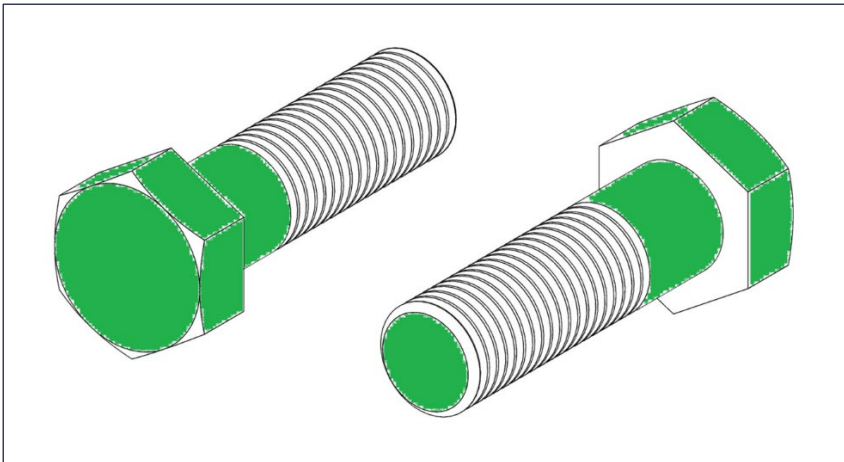


Figure 2 – Showing areas for measuring coating thickness (green) on Hexagon Headed Bolts and Screws

7.2. Pre-production (initial sample) inspection requirements for Hexagon Headed Bolts and Screws

7.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 2.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 1
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	Controlled Pandrol drawing (See Note)
Corrosion preventive coating	Coating thickness shall be in accordance with Table 1 in areas indicated in Figure 2	See Table 1 and Figure 2
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 2 – Pre-production (initial sample) approval inspection requirements

7.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

7.3. Bulk production inspection requirements for Hexagon Headed Bolts and Screws

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 3 shall be met during production of Hexagon Headed Bolts and Screws.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Bolts / Screws shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Bolts / Screws shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing (See Note)	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 1 in areas indicated in Figure 2	See Table 1 and Figure 2	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 3 – Production inspection requirements – Hexagon Headed Bolts and Screws

8. Masonry and Foundation Bolts (Splayed Anchors)

8.1. Requirements

Pandrol splayed anchors conform to the requirements of DIN 529:1984, Type D and Table 4.



Figure 3 – Splayed Anchor

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6g		
	International Standard	ISO 724, ISO 965-1		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 4 – Product requirements for Masonry and Foundation Bolts (Splayed Anchors)

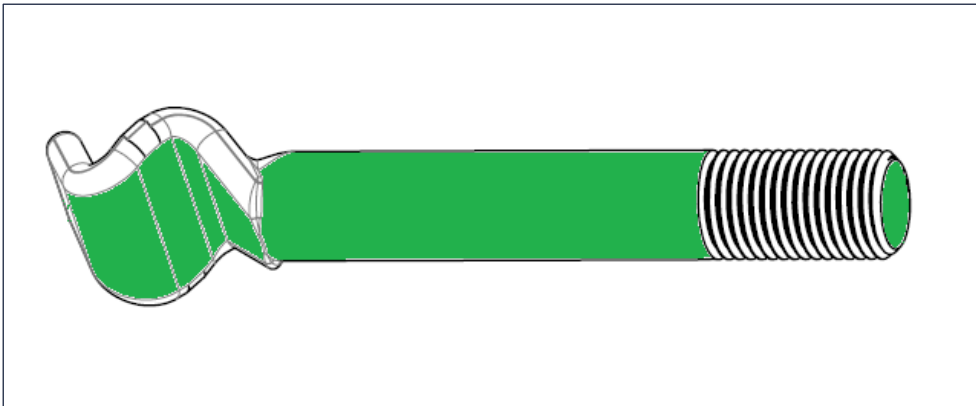


Figure 4 – Showing areas for measuring coating thickness (green) on Splayed Anchors

8.2. Pre-production (initial sample) inspection requirements for Splayed Anchors

8.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 5.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 4
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	See Table 4 and Figure 4
Corrosion preventive coating	Coating thickness shall be in accordance with Table 4 in areas indicated in Figure 4	
Dry assembly test	Coated splayed anchors shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 5 – Pre-production (initial sample) approval inspection requirements

8.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

8.3. Bulk production inspection requirements for Splayed Anchors

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 6 shall be met during production of Splayed Anchors.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Splayed anchors shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Splayed anchors shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing (See Note)	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 4 in areas indicated in Figure 4	See Table 4 and Figure 4	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated splayed anchors shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 6 – Production inspection requirements – Splayed Anchors

9. Square Headed Bolts and Screws

9.1. Requirements

Pandrol square headed bolts and screws conform to the requirements of BS 64 and Table 7.

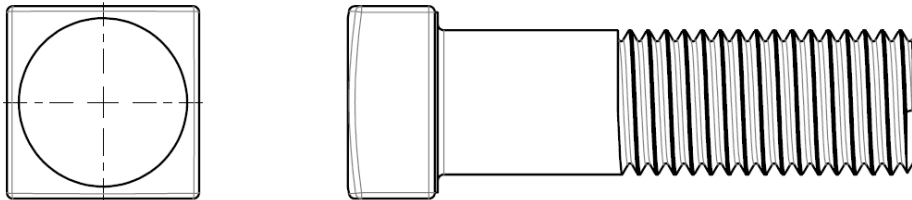


Figure 5 – Square Headed Bolt

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Thread	Tolerance	6g	
	International Standard	ISO 724, ISO 965-1	
Tolerance	Product grade	B	
	International Standard	ISO 4759-1	
Mechanical properties	International Standard	ISO 898-1	
	Property class	8.8	10.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 7 – Product requirements for Square Headed Bolts and Screws

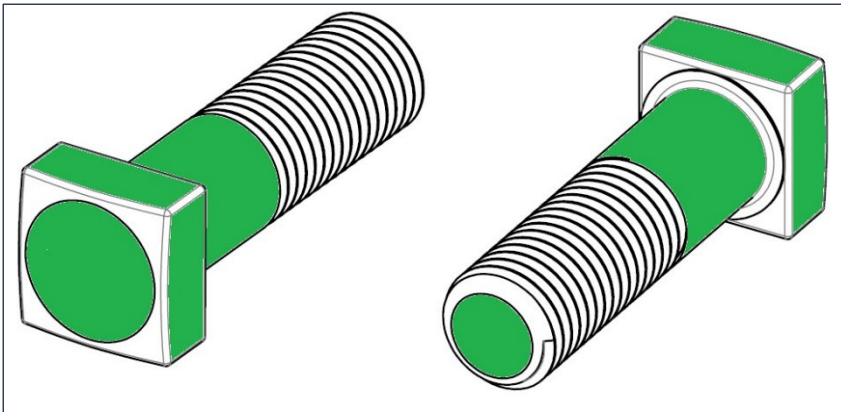


Figure 6 – Showing areas for measuring coating thickness (green) on Square Headed Bolts and Screws

9.2. Pre-production (initial sample) inspection requirements for Square Headed Bolts and Screws

9.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 8.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 7
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 7 in areas indicated in Figure 6	See Table 7 and Figure 6
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 8 – Pre-production (initial sample) approval inspection requirements

9.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

9.3. Bulk production inspection requirements for Square Headed Bolts and Screws

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 9 shall be met during production of Square Headed Bolts and Screws.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Bolts / Screws shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Bolts / Screws shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing (See Note)	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 7 in areas indicated in Figure 6	See Table 7 and Figure 6	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 9 – Production inspection requirements – Square Headed Bolts and Screws

10. Bolts with Trapezoidal Threads

10.1. Requirements

Pandrol bolts with trapezoidal threads conform to the requirements of NF F 50-014 and Table 10.

The head shape for bolts with trapezoidal threads may also be Hexagon Headed or Hexagon Socket Head Cap and reference should be made to the appropriate standards and tables in this document.

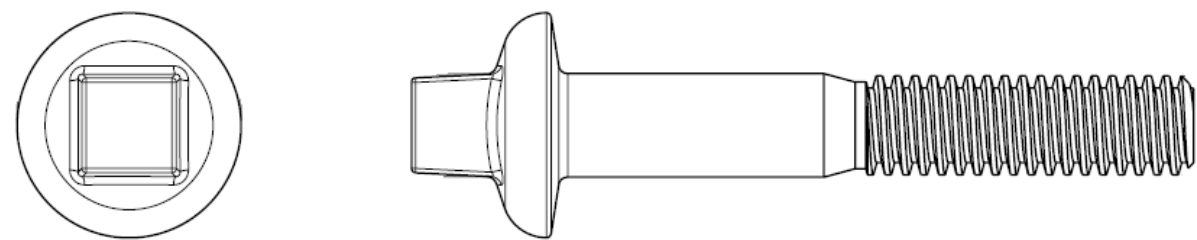


Figure 7 – Bolt with Trapezoidal Thread

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Thread	International Standards	ISO 2901, ISO 2902, ISO 2903	
Tolerance	Product grade	B	
	International Standard	ISO 4759-1	
Mechanical properties	International Standard	ISO 898-1	
	Property class	5.6	8.8
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 10 – Product requirements for Bolts with Trapezoidal Threads

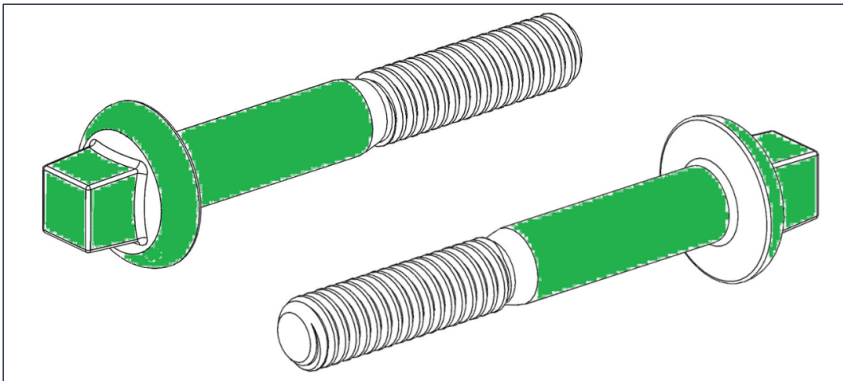


Figure 8 – Showing areas for measuring coating thickness (green) on Bolts with Trapezoidal Threads

10.2. Pre-production (initial sample) inspection requirements for Bolts with Trapezoidal Threads

10.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 11.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 10
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	Controlled Pandrol drawing (See Note)
Corrosion preventive coating	Coating thickness shall be in accordance with Table 10 in areas indicated in Figure 8	See Table 10 and Figure 8
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Bend test	The body of the Bolts shall be able to support bending of 15° without any ruptures, hot cracks, splits or incipient cracks. No rupture shall occur during bending up to 30°	Section 10.2.2
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 11 – Pre-production (initial sample) approval inspection requirements

10.2.2. Bend test requirements

The Bolt is placed horizontally, up to half its length, in a matrix with a smooth hole with diameter equal to a maximum of 1.1 times the diameter at the top of the threads.

Using a press, the part above the matrix is brought to an angle of 15° with the concealed part.

After visual inspection of the part, it is distorted again to an angle of 30°.

The body of the Bolts shall be able to support bending of 15° without any ruptures, hot cracks, splits or incipient cracks. The coating shall be scraped away to allow visual inspection if necessary.

No rupture shall occur during bending up to 30°.

- NOTE: This test replicates the test within NF F 50-014

10.2.3. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

10.3. Bulk production inspection requirements for Bolts with Trapezoidal Threads

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 12 shall be met during production of Bolts with Trapezoidal Threads.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Bolts shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Bolts shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 10 in areas indicated in Figure 8	See Table 10 and Figure 8	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Bend test	As defined in Section 10.2.2	Section 10.2.2	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated bolts shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 12 – Production inspection requirements – Bolts with Trapezoidal Threads

11. Double Ended Studs

11.1. Requirements

Pandrol double ended studs conform to the requirements of BS 4439 and Table 13.

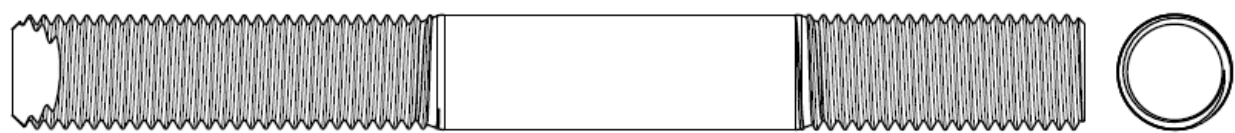


Figure 9 – Double Ended Stud

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6g		
	International Standard	ISO 724, ISO 965-1		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 13 – Product requirements for Double Ended Studs

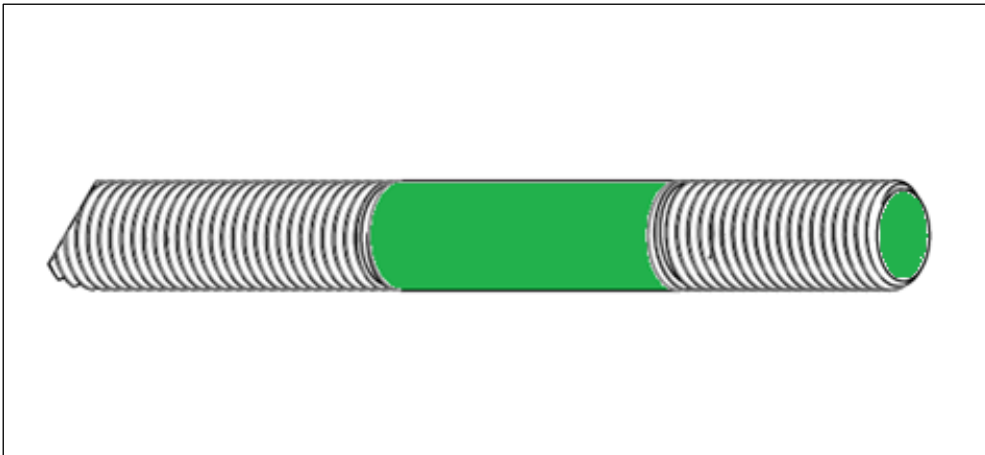


Figure 10 – Showing areas for measuring coating thickness (green) on Double Ended Studs

11.2. Pre-production (initial sample) inspection requirements for Double Ended Studs

11.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 14.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 13
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	See Table 13 and Figure 10
Corrosion preventive coating	Coating thickness shall be in accordance with Table 13 in areas indicated in Figure 10	
Dry assembly test	Coated right-hand threads of studs shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 14 – Pre-production (initial sample) approval inspection requirements

11.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

11.3. Bulk production inspection requirements for Double Ended Studs

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 15 shall be met during production of Double Ended Studs.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Studs shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Studs shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 13 in areas indicated in Figure 10	See Table 13 and Figure 10	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated right-hand threads of studs shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 15 – Production inspection requirements – Double Ended Studs

12.Hexagon socket head cap screws

12.1. Requirements

Pandrol Cap Screws conform to the requirements of ISO 4762 and Table 16.

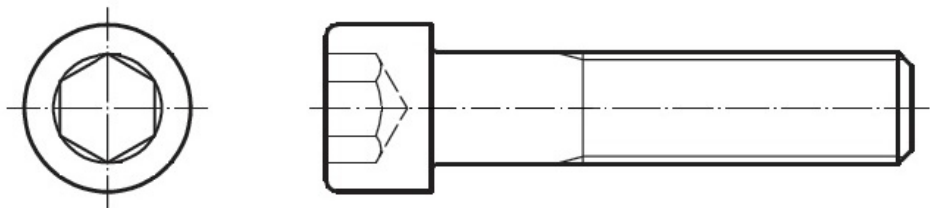


Figure 11 – Cap Screw

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6g		
	International Standard	ISO 724, ISO 965-1		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical properties	International Standard	ISO 898-1		
	Property class	8.8	10.9	12.9
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 16 – Product requirements for Hexagon socket head cap screws

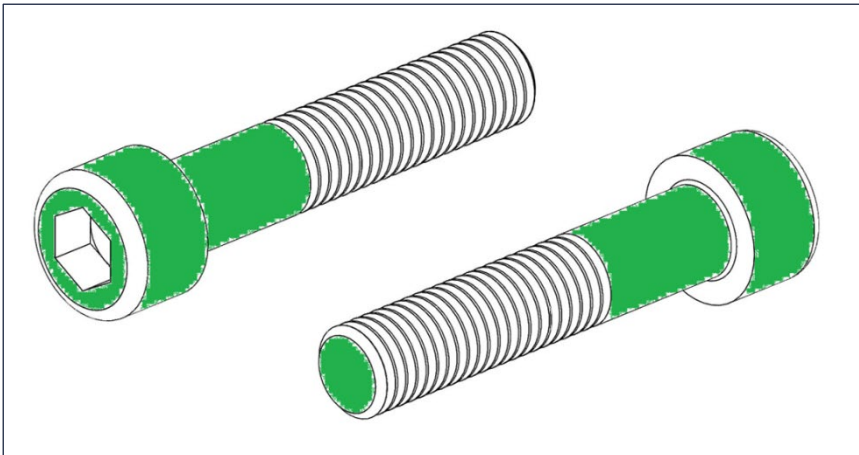


Figure 12 – Showing areas for measuring coating thickness (green) on Cap Screws

12.2. Pre-production (initial sample) inspection requirements for Cap Screws

12.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 17.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 16
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 16 in areas indicated in Figure 12	See Table 16 and Figure 12
Dry assembly test	Coated cap screws shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 17 – Pre-production (initial sample) approval inspection requirements

12.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

12.3. Bulk production inspection requirements for Cap Screws

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 18 shall be met during production of Cap Screws.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Cap Screws shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Cap Screws shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing (See Note)	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 16 in areas indicated in Figure 12	See Table 16 and Figure 12	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Coated cap screws shall be able to be assembled to a suitable female threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 18 – Production inspection requirements – Cap Screws

13. Screwspikes / Coachscrews

13.1. Requirements

Pandrol Screwspikes conform to the requirements of UIC 864-1 and Table 19.

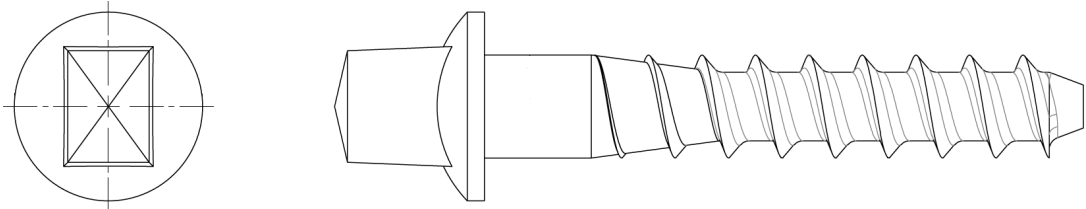


Figure 13 –Screwspike

- NOTE1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: The head shape can differ from standard, in accordance with the relevant Controlled Pandrol drawing.

Item		Specification / reference	
Material		Steel	
Tolerance	Product grade	B	
	International Standard	ISO 4759-1	
Mechanical properties	International Standard	ISO 898-1	
	Property class	5.6	8.8
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness	
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness	
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness	
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness	

Table 19 – Product requirements for Screwspikes

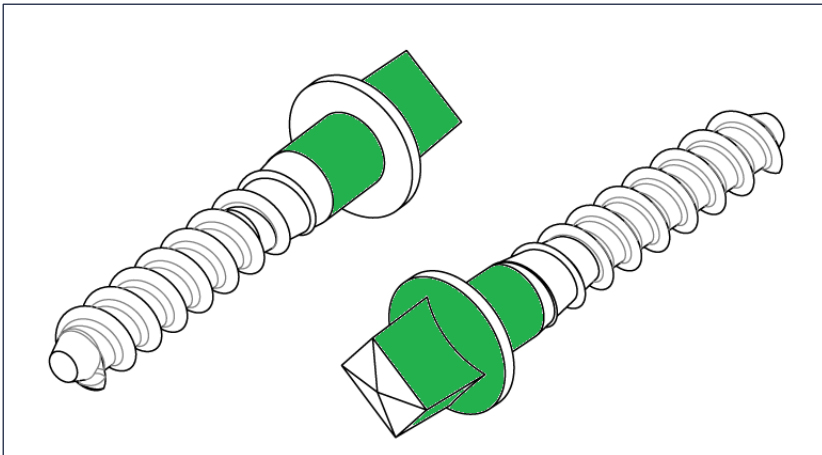


Figure 14 – Showing areas for measuring coating thickness (green) on Screwspikes

13.2. Pre-production (initial sample) inspection requirements for Screwspikes

13.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 20.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 19
UTS and elongation		
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing
	Threads (where specified) shall be clean, uniform and complete	
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing (See Note)
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 19 in areas indicated in Figure 14	See Table 19 and Figure 14
Bend test	No cracks, scaling or other faults shall occur when the Screwspike thread is bent as defined in Section 13.2.2	Section 13.2.2
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 20 – Pre-production (initial sample) approval inspection requirements

13.2.2. Bend test requirements

The threaded part of the Screwspike shall be bent to an angle of 15 degrees around a mandrel with a diameter equal to 4.d, where “d” is equal to the diameter of the Screwspike shank.

During this test, no cracks, scaling or other fault whatsoever shall occur.

- NOTE: This test replicates the test within UIC 864-1:1982

13.2.3. Pandrol’s inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

13.3. Bulk production inspection requirements for Screwspikes

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 21 shall be met during production of Screwspikes.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Screwspikes shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Screwspikes shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing (See Note)	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 19 in areas indicated in Figure 14	See Table 19 and Figure 14	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Bend test	As defined in Section 13.2.2	Section 13.2.2	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 21 – Production inspection requirements – Screwspikes

14. Hexagon plain nuts

14.1. Requirements

Pandrol hexagon plain nuts conform to the requirements of ISO 4032 and DIN 934 and Table 22.

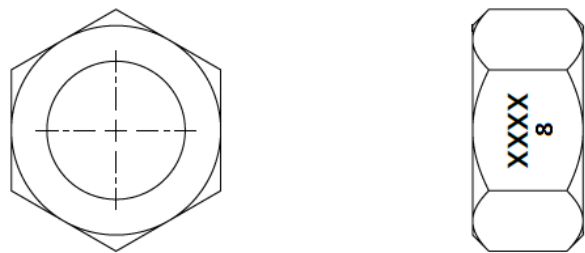


Figure 15 – Plain Nut

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel		
Thread	Tolerance	6AZ (for clearance of coated male threads)		
	International Standard	ISO 724, ISO 965-1, ISO 965-5		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical properties	International Standard	ISO 898-2, and LW-L Annex A for proof-load testing		
	Property class	6	8	10
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 22 – Product requirements for Hexagon plain nuts

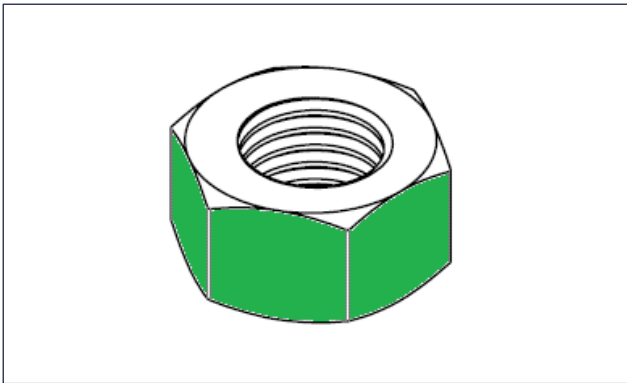


Figure 16 – Showing areas for measuring coating thickness (green) on Plain Nuts

14.2. Pre-production (initial sample) inspection requirements for Plain Nuts

14.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 23.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 22
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Markings (where specified)	Markings shall be present, legible and correct	
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	See Table 22 and Figure 16
Corrosion preventive coating	Coating thickness shall be in accordance with Table 22 in areas indicated in Figure 16	
Dry assembly test	Nuts shall be able to be assembled to a suitable coated male threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing
Weight	Weight of items shall be recorded for future production reference and commercial use.	

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 23 – Pre-production (initial sample) approval inspection requirements

14.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

14.3. Bulk production inspection requirements for Plain Nuts

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 24 shall be met during production of Plain Nuts.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Nuts shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Nuts shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 22 in areas indicated in Figure 16	See Table 22 and Figure 16	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dry assembly test	Nuts shall be able to be assembled to a suitable coated male threaded fastener, by-hand, ie without use of tools or lubrication	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 24 – Production inspection requirements – Plain Nuts

15. Prevailing Torque Hexagon Nuts with non-metallic insert (Nyloc Nuts)

15.1. Requirements

Pandrol Nyloc thin nuts conform to the requirements of ISO 10511 and DIN 985, and Table 25.

Pandrol Nyloc full nuts conform to the requirements of ISO 7040, and Table 25.

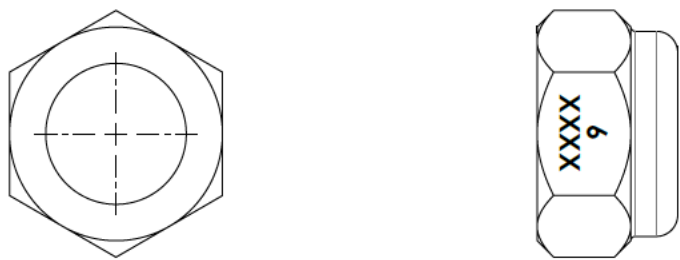


Figure 17 – Nyloc Nut

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference		
Material		Steel and Polyamide		
Thread	Tolerance	6AZ (for clearance of coated male threads)		
	International Standard	ISO 724, ISO 965-1, ISO 965-5		
Tolerance	Product grade	B		
	International Standard	ISO 4759-1		
Mechanical and performance properties	International Standard	ISO 898-2, ISO 2320 and LW-L Annex A for proof-load testing		
	Property class	6	8	10
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness		
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness		

Table 25 – Product requirements for Prevailing Torque Hexagon Nuts with non-metallic insert (Nyloc Nuts)

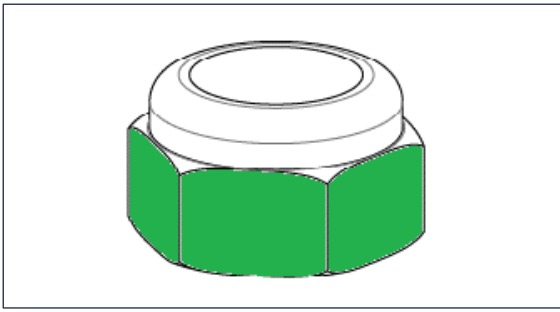


Figure 18 – Showing areas for measuring coating thickness (green) on Nyloc Nuts

15.2. Pre-production (initial sample) inspection requirements for Nyloc Nuts

15.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 26.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 25
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing (See Note)
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 25 in areas indicated in Figure 18	See Table 25 and Figure 18
Prevailing torque test	Nyloc nuts shall meet the prevailing torque test requirements for 1 st installation, 1 st removal and 5 th removal	ISO 2320
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 26 – Pre-production (initial sample) approval inspection requirements

15.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

15.3. Bulk production inspection requirements for Nyloc Nuts

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 27 shall be met during production of Nyloc Nuts.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Nuts shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Markings (where specified)	Markings shall be present, legible and correct	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Nuts shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 25 in areas indicated in Figure 18	See Table 25 and Figure 18	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Prevailing torque test	Nyloc nuts shall meet the prevailing torque test requirements for 1 st installation, 1 st removal and 5 th removal	ISO 2320	3 samples at start-up and 1 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 27 – Production inspection requirements – Nyloc Nuts

16. Plain washers

16.1. Requirements

Pandrol plain washers conform to the requirements of BS 4320 and Table 28.

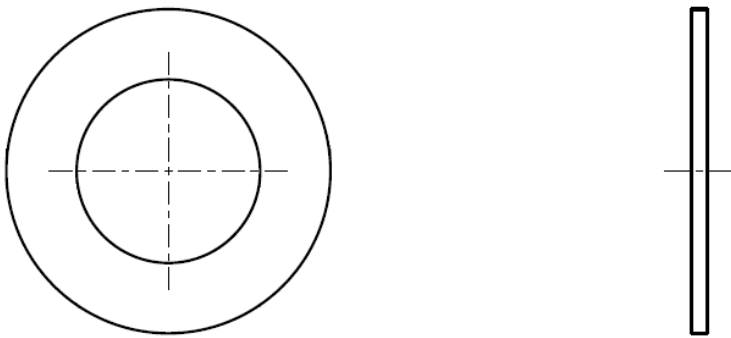


Figure 19 – Plain Washer

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Item		Specification / reference
Material		Steel as specified in BS 4320 (Cold Rolled Strip CS4 in the hard condition)
		Steel # 45 as specified in China Standard GB/T 699
Coating	Corrosion protection	Hot dip galvanised to ISO 10684, 50µm batch average thickness
	Alternative corrosion protection	Sherardized to ISO 17668, Class 30, 30µm minimum local thickness
		Zinc Flake coating to ISO 10683, 5 to 50µm thickness
		Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness

Table 28 – Product requirements for Plain Washers

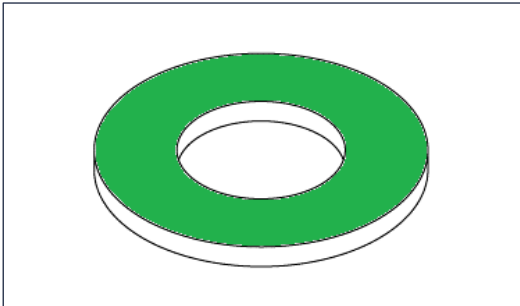


Figure 20 – Showing areas for measuring coating thickness (green) on Plain Washers

16.2. Pre-production (initial sample) inspection requirements for Plain Washers

16.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 29.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 28
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing (See Note)
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 28 in areas indicated in Figure 20	See Table 28 and Figure 20
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 29 – Pre-production (initial sample) approval inspection requirements

16.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

16.3. Bulk production inspection requirements for Plain Washers

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 30 shall be met during production of Plain Washers.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Washers shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Washers shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 28 in areas indicated in Figure 20	See Table 28 and Figure 20	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 30 – Production inspection requirements – Plain Washers

17. Disc Springs

17.1. Requirements

Pandrol disc springs conform to the requirements of EN 16983 and Table 31.

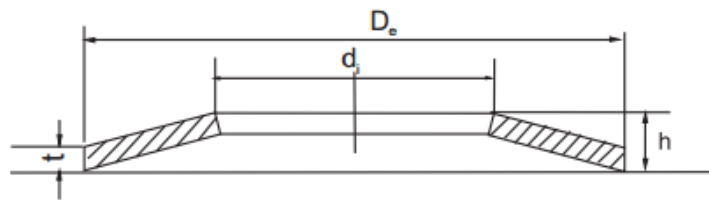


Figure 21 – Disc Spring

- NOTE 1: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.
- NOTE 2: EN 16983 has replaced DIN 2093

Item		Specification / reference
Material		Spring steel
	Hydrogen embrittlement test	Parallel bearing surface test to ISO 15330
Coating	Corrosion protection	Mechanical zinc plate with iridescent / yellow chromate conversion to ISO 12683, 12µm minimum local thickness

Table 31 – Product requirements for Disc Springs

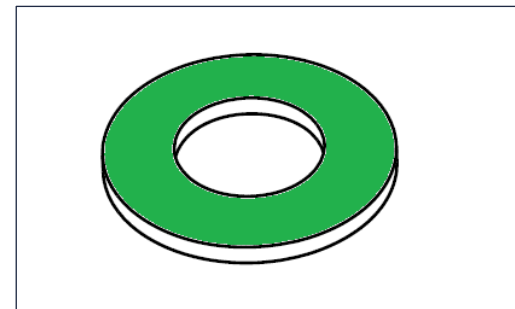


Figure 22 – Showing areas for measuring coating thickness (green) on Disc Springs

17.2. Pre-production (initial sample) inspection requirements for Disc Springs

17.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 32.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 31
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	
Corrosion preventive coating	Coating thickness shall be in accordance with Table 31 in areas indicated in Figure 22	See Table 31 and Figure 22
Hydrogen embrittlement test (Uncoated Disc Spring washers only)	As-made Disc Springs shall be torque tightened to 200Nm and shall not fracture during the test of 72 hours duration. Test shall start within 24 hours of the last process of the as-made washers.	ISO 15330
Hydrogen embrittlement test (Coated Disc Spring washers only)	Coated Disc Springs shall be torque tightened to 200Nm and shall not fracture during the test of 72 hours duration. Test shall start within 24 hours of the last process of the as-made washers.	ISO 15330
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 32 – Pre-production (initial sample) approval inspection requirements

17.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

17.3. Bulk production inspection requirements for Disc Springs

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 33 shall be met during production of Disc Springs.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Disc Springs shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Disc Springs shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Hydrogen embrittlement test	As-made Disc Springs shall be torque tightened to 200Nm and shall not fracture during the test of 72 hours duration. Test shall start within 24 hours of the last process of the as-made washers.	ISO 15330	16 samples at start-up and 16 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4
	Coated Disc Springs shall be torque tightened to 200Nm and shall not fracture during the test of 72 hours duration. Test shall start within 24 hours of the last process of the as-made washers.	ISO 15330	16 samples at start-up and 16 additional for every 10000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 31 in areas indicated in Figure 22	See Table 31 and Figure 22	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 33 – Production inspection requirements – Disc Springs

18. Bonded seal washers

18.1. Requirements

Pandrol bonded seal washers conform to the requirements of the Pandrol drawing and Table 34.

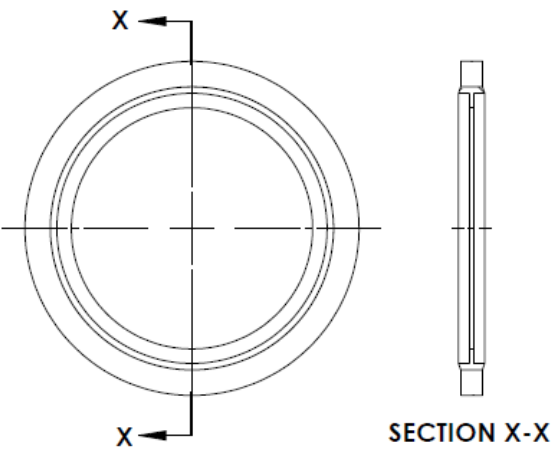


Figure 23 – Bonded Seal Washer

- NOTE: Dimensions and tolerances specified on the relevant Controlled Pandrol drawing shall be used.

Item		Specification / reference
Material		Steel and Nitrile Rubber
Coating	Corrosion protection	Electroplated zinc with iridescent / yellow chromate conversion to ISO 4042, 12µm minimum local thickness
	Alternative corrosion protection	Mechanical zinc plate and iridescent / yellow chromate conversion to ISO 12683, 12µm minimum local thickness

Table 34 – Product requirements for Bonded Seal Washers

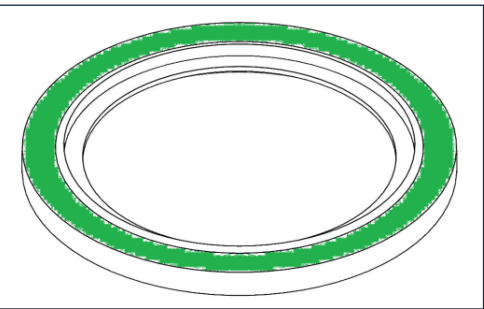


Figure 24 – Showing areas for measuring coating thickness (green) on Bonded Seal Washers

18.2. Pre-production (initial sample) inspection requirements for Bonded Seal Washers

18.2.1. Manufacturer's inspection

Prior to submitting any samples, the manufacturer shall ensure their products meet all requirements of this specification by testing three samples from each tool in accordance with Table 35.

Test results shall be treated as though they were independently obtained for each property.

Item	Specification	Reference
Raw material chemistry	Shall be in accordance with the raw material standards in the relevant product requirements table	See Table 34
Hardness		
Visual Inspection	Items shall be clean and free of burrs and sharp edges Threads (where specified) shall be clean, uniform and complete	Controlled Pandrol drawing
Dimensions	All dimensions of 'as made' items (before the addition of any coating) shall be in accordance with the appropriate Controlled Pandrol drawing	Controlled Pandrol drawing (See Note)
Corrosion preventive coating	Coating thickness shall be in accordance with Table 34 in areas indicated in Figure 24	See Table 34 and Figure 24
Weight	Weight of items shall be recorded for future production reference and commercial use.	-

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 35 – Pre-production (initial sample) approval inspection requirements

18.2.2. Pandrol's inspection

Where compliance with all requirements in this specification is found, the manufacturer shall submit to Pandrol:

- The pre-production (initial sample) inspection reports
- All tested samples
- Three untested samples from each tool

All samples submitted shall be appropriately packaged, sealed and labelled to suitably identify their contents.

18.3. Bulk production inspection requirements for Bonded Seal Washers

Unless otherwise agreed in writing by Pandrol, the specifications shown in Table 36 shall be met during production of Bonded Seal Washers.

Item	Specification	Reference	Inspection frequency	Action if failed
Visual Inspection	Washers shall be clean and free of burrs and sharp edges	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Dimensions (see Note)	Washers shall comply with bulk production dimensions shown on relevant Pandrol drawing.	Controlled Pandrol drawing	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4
Corrosion protective coating	Coating thickness shall be in accordance with Table 34 in areas indicated in Figure 24	See Table 34 and Figure 24	3 samples at start-up and 1 additional for every 5000	Quarantine the affected quantity and refer to Section 4.4

- NOTE: Dimensions and tolerances shall comply with the Controlled Pandrol drawing. Any dimensions or tolerances not specified on the Pandrol drawing shall be taken from the manufacturing standard referenced. If a conflict exists, the Controlled Pandrol drawing shall be used.

Table 36 – Production inspection requirements – Bonded Seal Washers

Annexes

A. Proof load testing of Nuts with thread fit tolerance 6AZ

Background

The proof load test for nuts specified in ISO 898-2 is used to confirm that the nut has the required mechanical performance properties to prevent thread stripping. The threaded interface between the nut and the test mandrel is critical to ensuring that the load can be fully transmitted between the test mandrel and nut and for this reason, the tolerances for the test mandrel are prescribed in ISO 898-2.

The ISO 898-2 proof load test method requires that the thread fit tolerance for the test mandrel shown in Figure A.1 shall be precisely controlled with clear definition of pitch diameter and major diameter. However, the dimensions prescribed for this test mandrel are designed to be used with 'standard' thread fit and are not compatible with relatively thick corrosion protective coatings such as Hot-Dip Galvanising, which require nut threads to be tapped oversize. Such oversize thread fits are used by Pandrol.

The result of proof load testing nuts with an oversize 6AZ thread fit tolerance on a 'standard' test mandrel will be a much smaller interface between the screw threads, most likely resulting in a much-reduced proof-load and normally failing to meet the minimum requirements specified in ISO 898-2.

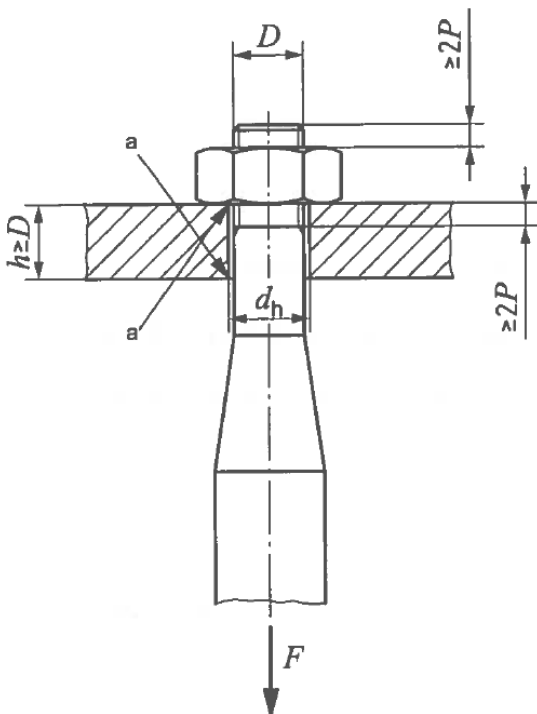


Figure A. 1 – Proof load test diagram, from ISO 898-2

Proof Load Tests

Proof load testing of nuts with a 6AZ thread fit, as specified in this Pandrol Product Specification document shall be performed in accordance with the requirements of ISO 898-2, but using a mandrel which has been manufactured with over-sized threads to suit the offset 6AZ thread fit. The offset used to define the thread pitch diameter dimensions and external thread dimensions for the test mandrel shall be the same offset as defined in ISO 965-5 for the appropriate thread size.

The example below illustrates that the test mandrel for proof load testing M24 nuts with a 6AZ thread fit should be 0.360mm larger than the dimensions stated in ISO 898-2.

e.g. Test mandrel dimensions for M24 x 3, 6AZ thread fit comparison

Extracted sizes taken from ISO 898-2:2012 for test mandrel				
	External thread diameter of the mandrel (lowest quarter of tolerance class 6g)		Pitch diameter of the mandrel (tolerance class 5h)	
	Max	Min	Max	Min
M24 x 3	23.671	23.577	22.051	21.891

Table A. 1 – Extracted sizes taken from ISO 898-2:2012

Comparison of Nuts with 6H to 6AZ thread fit, taken from ISO 965-2 and ISO 965-5				
	Pitch diameter		Minor diameter	
	Max	Min	Max	Min
M24 x 3 (6H)	22.316	22.051	21.252	20.752
M24 x 3 (6AZ)	22.676	22.411	21.612	21.112
Offset	0.360	0.360	0.360	0.360

Table A. 2 – Comparison of Nuts with 6H to 6AZ thread fit, taken from ISO 965-2 and ISO 965-5

Resultant test mandrel dimensions				
	External thread diameter of the mandrel (lowest quarter of tolerance class 6g + offset)		Pitch diameter of the mandrel (tolerance class 5h + offset)	
	Max	Min	Max	Min
Mandrel dimensions including 6AZ offset	24.031	23.937	22.411	22.251

Table A. 3 – Resultant test mandrel dimensions

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[pandrol.com](https://www.pandrol.com)

Pandrol Rail Fastening Systems
Gateford Road,
Worksop,
Nottinghamshire
S81 7AX
UK

+44 (0)1909 476101
www.pandrol.com

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